Shahnawaz A. Adil, Ph.D.

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Current Position

2024 – Present **Postdoctoral Fellow**, Instituto de Ciencias Físicas (ICF), UNAM, Mexico.
Working on dark energy, cosmological tensions, and large-scale structure
Advisor: Dr. José Alberto Vázquez González

Research Interests

- **Cosmology**: Dark Energy, Large Scale Structure, Hubble tension, S_8 tension, Modified gravity, dark matter–dark energy interaction, scalar fields.

Education

2018 – 2024 **Ph.D. in Physics**, Jamia Millia Islamia, Delhi, India
Thesis: *Cosmological non-standard field dynamics in light of observations*
Advisor: Dr. Somasri Sen

2014 – 2016 **M.Sc. in Physics**, Jamia Millia Islamia, Delhi, India
CGPA: 8.04/10 (Distinction)
Thesis: *Poincaré–Cartan forms for extended phase space in curved background*
Supervisor: Prof. Pankaj Sharan

2011 – 2014 **B.Sc. (Hons.) Physics**, Jamia Millia Islamia, Delhi, India
First Division (68%)

Publications

Journal Articles

1. **S. A. Adil**, M. A. Zapata, Ö. Akarsu, and J. A. Vazquez, "Background-level reconstruction of scalar-field potentials from dark-energy histories and comparison with analytic potential families," Mar. 2026. arXiv: 2603.14693 [astro-ph.CO].

2. D. Kumar, A. Mitra, **S. A. Adil**, and A. A. Sen, "Exploring alternative cosmologies with the LSST: Simulated forecasts and current observational constraints," *Phys. Rev. D*, vol. 111, no. 4, p. 043 503, 2025. DOI: 10.1103/PhysRevD.111.043503. arXiv: 2406.06757 [astro-ph.CO].

3. E. Specogna et al., "Updated Constraints on Omnipotent Dark Energy: A Comprehensive Analysis with CMB and BAO Data," Apr. 2025. arXiv: 2504.17859 [gr-qc].

4. M. A. Zapata, G. Garcia-Arroyo, **S. A. Adil**, and J. A. Vazquez, "How Holographic is the Dark Energy? A Spline Nodal reconstruction approach," Jul. 2025. arXiv: 2507.22292 [astro-ph.CO].

5. **S. A. Adil**, M. G. Dainotti, and A. A. Sen, "Revisiting the concordance ΛCDM model using Gamma-Ray Bursts together with supernovae Ia and Planck data," *JCAP*, vol. 08, p. 015, 2024. DOI: 10.1088/1475-7516/2024/08/015. arXiv: 2405.01452 [astro-ph.HE].

6 **S. A. Adil** et al., "Omnipotent dark energy: A phenomenological answer to the Hubble tension," *Phys. Rev. D*, vol. 109, no. 2, p. 023 527, 2024. DOI: 10.1103/PhysRevD.109.023527. arXiv: 2306.08046 [astro-ph.CO].

7 N. Menci, **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, "Negative cosmological constant in the dark energy sector: tests from JWST photometric and spectroscopic observations of high-redshift galaxies," *JCAP*, vol. 07, p. 072, 2024. DOI: 10.1088/1475-7516/2024/07/072. arXiv: 2401.12659 [astro-ph.CO].

8 **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, "Dark energy in light of the early JWST observations: case for a negative cosmological constant?" *JCAP*, vol. 10, p. 072, 2023. DOI: 10.1088/1475-7516/2023/10/072. arXiv: 2307.12763 [astro-ph.CO].

9 **S. A. Adil** et al., "S8 increases with effective redshift in ΛCDM cosmology," *Mon. Not. Roy. Astron. Soc.*, vol. 528, no. 1, pp. L20–L26, 2023. DOI: 10.1093/mnrasl/slad165. arXiv: 2303.06928 [astro-ph.CO].

10 A. Bassi, **S. A. Adil**, M. P. Rajvanshi, and A. A. Sen, "Cosmological evolution in bimetric gravity: observational constraints and LSS signatures," *Eur. Phys. J. C*, vol. 83, no. 6, p. 525, 2023. DOI: 10.1140/epjc/s10052-023-11707-4. arXiv: 2301.11000 [astro-ph.CO].

11 Ruchika, **S. A. Adil**, K. Dutta, A. Mukherjee, and A. A. Sen, "Observational constraints on axion(s) dark energy with a cosmological constant," *Phys. Dark Univ.*, vol. 40, p. 101 199, 2023. DOI: 10.1016/j.dark.2023.101199. arXiv: 2005.08813 [astro-ph.CO].

12 A. A. Sen, **S. A. Adil**, and S. Sen, "Do cosmological observations allow a negative Λ?" *Mon. Not. Roy. Astron. Soc.*, vol. 518, no. 1, pp. 1098–1105, 2022. DOI: 10.1093/mnras/stac2796. arXiv: 2112.10641 [astro-ph.CO].

13 **S. A. Adil**, M. R. Gangopadhyay, M. Sami, and M. K. Sharma, "Late-time acceleration due to a generic modification of gravity and the Hubble tension," *Phys. Rev. D*, vol. 104, no. 10, p. 103 534, 2021. DOI: 10.1103/PhysRevD.104.103534. arXiv: 2106.03093 [astro-ph.CO].

Skills

Languages	English (fluent), Spanish (poquito), Hindi, Urdu
Coding	Python, Bash, LaTeX
Scientific Tools	CLASS, CAMB, MontePython, Cobaya, SIMPLEMC, GetDist
Inference Methods	Emcee, MultiNest, PolyChord, George, GaPP
Computing	Linux, Windows, HPC environments, MPI parallelization

Conferences

Nov 2024	Presented a talk at *IX Essential Cosmology for the Next Generation*, Playa del Carmen, Mexico, on "Dark Energy in the Light of JWST Anomalies".
	Presented a colloquium talk at *Instituto de Ciencias Físicas (ICF), UNAM*, on "13.8 billion years in the making: the evolution of our Universe and its unsolved mysteries".
Aug 2024	Presented a talk at *21st Century Cosmology: Tensions, Anomalies & New Physics*, Ashoka University, India, on "Cosmological tensions and new physics in the dark energy sector".
Dec 2022	Presented a talk at *Indian Association for General Relativity and Gravitation (IAGRG)*, on "Is there an evidence of negative cosmological constant through recent cosmological observations?".
Nov 2022	Presented an e-poster at *Young Astronomers' Meet*, ARIES, Nainital, titled "Evidence of an AdS vacua in the Universe".

Conferences (continued)

Oct 2022	Presented a talk at *Cosmology@CCSP_1*, on "Late-time acceleration due to a generic modification of gravity and Hubble tension".
Nov 2021	Presented a talk at *Midwest Relativity Meeting*, on "Late-time acceleration due to a generic modification of gravity and Hubble tension".

Outreach & Mentorship

2024	Taught a cosmology program to school children (link).
2023	Co-mentored Dharmendra Kumar for Ph.D. course work.
2022	Co-mentored Amresh Verma for M.Sc. dissertation.
2021	Co-mentored Sonej Alam for Ph.D. course work.
	Volunteered for science outreach program for **Square Kilometre Array (SKA)** in Vigyan Samagam.

Schools & Seminars

Sept 2024	Regular participant in weekly seminars at **Instituto de Ciencias Físicas (ICF), UNAM**.
July 2023	Attended *Summer School on Cosmology*, ICTP.
July 2022	Attended *Summer School on Cosmology*, ICTP.
Dec 2020	Attended the 31st meeting of **IAGRG**, IIT Gandhinagar.
Sept 2020	Attended *Physics of the Early Universe*, ICTS (online).
Nov 2019	Attended **SERB Preparatory School in Theoretical High Energy Physics (THEP)**, Tezpur University.

Teaching Experience

Feb 2025	Teaching Assistant for the course "Machine Learning and Cosmology" (Instructor: Dr. José Alberto Vázquez González).
Mar 2023	Taught physics to NEET/JEE aspirants at Orchid Institute, New Delhi (link).
Aug 2021	Conducted B.Sc. Physics laboratory sessions, Department of Physics.

References

Prof. Somasri Sen
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Prof. Eleonora Di Valentino
University of Sheffield, UK
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Dr. Sunny Vagnozzi
University of Trento, Italy
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Dr. J. Alberto Vázquez González
Researcher
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Prof. Anjan A. Sen
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